

Design and Implementation of an Online Dermatology Platform with Integrated Online Pharmacy

Medical Platform Project Team

January 20, 2026

Acknowledgement

First and foremost, we would like to express our deepest gratitude to our project advisor for their invaluable guidance, encouragement, and insightful feedback throughout the duration of this project. Their expertise and dedication have been instrumental in the successful completion of this work.

We are also grateful to the faculty and staff of the department for providing a conducive environment for learning and research. Special thanks go to the healthcare professionals, including dermatologists and pharmacists, who provided their expertise and perspective during the requirement gathering and analysis phases.

Finally, we would like to thank our family and friends for their unwavering support, patience, and motivation. Their belief in our abilities has been a constant source of strength.

Acronyms and Abbreviations

- **API:** Application Programming Interface
- **DB:** Database
- **JWT:** JSON Web Token
- **UI/UX:** User Interface / User Experience
- **HTTP:** Hypertext Transfer Protocol

Operational Definitions

- **Pharmacist:** A professional responsible for managing and dispensing medicine.
- **Dermatologist:** A doctor specializing in skin, hair, and nail health.
- **Patient:** A user seeking dermatology consultation or medicine.

List of Figures

2.1	Overall Use-Case Diagram	13
2.2	Patient Use Case Diagram	14
2.3	Doctor Use Case Diagram	14
2.4	Administrator Use Case Diagram	15
3.1	Sequence Diagram: Appointment Booking Flow	23
3.2	Sequence Diagram: Pharmacy Order Flow	24
3.3	Entity-Relationship Diagram of the System	25

Abstract

The integration of digital technology in healthcare has transformed how dermatology services and pharmacy operations are managed. This project proposes an "Online Dermatology Platform with an Integrated Online Pharmacy" to streamline patient care, from diagnosis to medicine procurement. The platform allows patients to register online, select specialized dermatology services, book appointments, and pay registration fees. Additionally, it features an online pharmacy module where patients can browse and purchase medicines, facilitating a complete healthcare cycle. Developed using Next.js, Node.js, and PostgreSQL, the system ensures secure data management and efficient communication between patients, dermatologists, pharmacists, and administrators. By automating registration, reminders, and payments, the system reduces administrative overhead and improves accessibility to especializados skincare and medication.

Contents

Acknowledgement	i
Acronyms and Abbreviations	i
Operational Definitions	i
List of Figures	ii
1 Introduction	1
1.1 Introduction	1
1.1.1 Background	1
1.1.2 Project Description	2
1.1.3 Statement of the Problem	2
1.1.4 Objectives of the Project	3
1.1.5 Feasibility Study	3
1.1.6 Significance of the Project	5
1.1.7 Beneficiaries of the Project	5
1.1.8 Scope of the Project	5
1.1.9 Methodology	5
1.1.10 Project Team and Communication	5
1.1.11 Development Tools and Technologies	5
1.1.12 Testing and Deployment Environment	6
1.1.13 Constraints and Assumptions	6
1.1.14 Risks	6
1.1.15 Project Phases and Deliverables	6
1.1.16 Work Breakdown Structure	6
1.1.17 Project Schedule and Timeline	6
2 Business Area Analysis and Requirement Envisioning	7
2.1 Introduction	7
2.1.1 Agile Analysis Approach	7
2.1.2 Stakeholder Analysis	8
2.1.3 Players of the Existing System (Roles)	8
2.1.4 Current Activities and Workflow Analysis	8

2.1.5	Forms and Reports of the Existing System	9
2.1.6	SWOT Analysis of the Existing System	10
2.1.7	Problems of the Current System	10
2.2	Requirements Envisioning	11
2.2.1	Non-Functional Requirements	11
2.2.2	Functional Requirements	11
2.2.3	System Requirements	12
2.3	Essential Use Cases	12
2.3.1	Actor Specification	12
2.3.2	Use Case Diagrams	13
2.4	Domain Modeling	15
2.5	Architectural Envisioning	15
2.5.1	System Architecture	15
2.5.2	Deployment Diagram	15
2.5.3	Persistence Data Management	16
2.5.4	Access Control and Security	16
3	Design and Construction	17
3.1	Introduction	17
3.2	Actor Identification	17
3.3	Business Rules	18
3.4	Sequence Diagrams	18
3.4.1	Appointment Booking Flow	18
3.4.2	Pharmacy Order Flow	19
3.5	System Decomposition	19
3.5.1	Patient Module	19
3.5.2	Clinical Module	19
3.5.3	Pharmacy Inventory Module	20
3.5.4	Payment and Billing Module	20
3.6	Data Modeling	20
3.6.1	Entity-Relationship (ER) Diagram	20
3.6.2	Class Diagrams	21
3.6.3	Component Diagram	21
3.7	User Interface Design	21
3.7.1	Design Principles	21
3.7.2	Wireframe Overview	22
3.8	Security Considerations	22
3.9	Design Rationale and Justification	22

4	Conclusion and Recommendations	26
4.1	Summary of Findings	26
4.2	Achievement of Objectives	26
4.3	Limitations	27
4.4	Recommendations and Future Work	27
5	References	28
6	Appendix	30
6.1	Appendix A: User Interview Guide	30
6.2	Appendix B: Database Schema (Detailed)	30
6.3	Appendix C: Project Timeline	31
6.4	Appendix D: User Manual Fragments	31

Chapter 1

Introduction

1.1 Introduction

The healthcare sector is undergoing a profound transformation driven by the rapid advancement of digital technologies and increased internet penetration. Web-based healthcare platforms have emerged as powerful tools for improving service delivery, enhancing patient engagement, and reducing operational inefficiencies. These systems enable healthcare providers to move beyond traditional, paper-based workflows and adopt data-driven, patient-centered models of care.

Dermatology, as a visually intensive medical specialty, is particularly well-suited for digital transformation. Many dermatological conditions can be assessed through clinical images, patient-reported symptoms, and medical history without requiring immediate physical examination. At the same time, access to dermatologists remains limited in many developing regions, including Ethiopia, where specialist care is often centralized in urban areas.

This project focuses on the intersection of dermatology and pharmacy services by proposing a unified digital platform that integrates online dermatological consultation with an online pharmacy. The goal is to provide an end-to-end healthcare solution that supports patients from initial consultation through diagnosis, prescription, and medication procurement. By leveraging modern web technologies, the platform aims to improve accessibility, efficiency, transparency, and quality of dermatological care.

1.1.1 Background

Healthcare systems in developing countries face persistent challenges related to limited infrastructure, workforce shortages, and inefficient service delivery mechanisms. Specialized medical services such as dermatology are often unavailable outside major cities, forcing patients to travel long distances for consultation and treatment. This situation disproportionately affects rural populations, leading to delayed diagnosis, increased financial

burden, and reduced treatment adherence.

In Ethiopia, dermatological services are typically delivered through hospital outpatient departments and private clinics, many of which rely on manual record-keeping and in-person consultations. After receiving a prescription, patients must independently locate pharmacies, often encountering issues such as medication unavailability, long queues, or counterfeit drugs. This fragmented approach to healthcare delivery results in inefficiencies, duplication of effort, and poor continuity of care.

Recent advancements in telemedicine, cloud computing, and secure online payment systems provide an opportunity to address these challenges. Teledermatology enables remote consultations through image-based diagnosis and asynchronous communication, while online pharmacy systems streamline prescription fulfillment and inventory management. Integrating these components into a single platform can significantly improve the patient journey and overall healthcare outcomes.

1.1.2 Project Description

The proposed system is a comprehensive web-based platform that integrates an online dermatology clinic with an online pharmacy. The platform is designed to support the entire patient care lifecycle, beginning with user registration and continuing through consultation, prescription issuance, payment processing, and medication acquisition.

Patients can create secure accounts, upload dermatological images, describe symptoms, and book appointments with licensed dermatologists. Dermatologists can review patient submissions, provide diagnoses, conduct follow-up consultations, and issue digital prescriptions. These prescriptions are seamlessly linked to the integrated pharmacy module, allowing patients to view medication availability, pricing, and order status in real time.

The system also includes administrative features such as appointment management, inventory tracking, payment reporting, and user role management. Built using Next.js, Node.js, and PostgreSQL, the platform emphasizes scalability, security, and performance while ensuring compliance with healthcare data protection standards.

1.1.3 Statement of the Problem

Traditional dermatology clinics and pharmacies often operate as independent entities, relying heavily on manual and paper-based processes. This lack of integration leads to multiple inefficiencies that negatively impact both patients and healthcare providers.

Key problems include:

- Difficulty for patients to locate and book appointments with qualified dermatologists, particularly in non-urban areas.

- High administrative workload associated with manual patient registration, record keeping, and billing.
- Absence of a centralized digital system for tracking prescriptions, medicine availability, and payments.
- Ineffective communication mechanisms, resulting in missed appointments and delayed prescription fulfillment.

These challenges contribute to reduced service quality, increased operational costs, and limited scalability of dermatology services. Addressing these issues requires a unified digital solution that connects clinical and pharmaceutical workflows.

1.1.4 Objectives of the Project

General Objective

The general objective of this project is to design and implement a secure, scalable, and user-friendly web-based platform that integrates online dermatology consultations with an online pharmacy to improve healthcare accessibility, efficiency, and service quality.

Specific Objectives

- To enable patients to register, authenticate, and manage their profiles online.
- To provide an appointment booking system for selecting dermatology services and available time slots.
- To facilitate remote dermatological consultations through image uploads and digital communication.
- To implement a secure payment gateway for consultation fees and medicine purchases.
- To develop an online pharmacy module for medicine inventory management and sales.
- To automate appointment reminders and prescription notifications.

1.1.5 Feasibility Study

Technical Feasibility

The proposed system leverages a modern and proven technology stack. Next.js provides a fast, responsive, and SEO-friendly frontend, while Node.js and Express enable scalable

backend services. PostgreSQL ensures reliable and consistent data storage. The system supports JWT-based authentication, encrypted data transmission, and integration with third-party services such as payment gateways and cloud storage.

The availability of these technologies, along with extensive documentation and developer communities, makes the project technically feasible within the given constraints.

Economic Feasibility

The economic feasibility of the project is supported by relatively low development and deployment costs compared to traditional healthcare infrastructure investments. Cloud-based hosting reduces upfront hardware expenses, while open-source tools minimize licensing costs.

Long-term benefits include:

- Reduced administrative and operational expenses.
- Increased revenue through online consultations and pharmacy sales.
- Improved cost efficiency through automation and digital workflows.

Schedule Feasibility

The project is planned for completion within a 12-week timeframe using an Agile development approach. The use of iterative development and continuous feedback ensures timely delivery of core functionalities while allowing flexibility for refinement and optimization.

Operational Feasibility

The platform is designed to align with existing clinical and pharmacy workflows. Minimal training is required due to the intuitive user interface, making adoption feasible for healthcare professionals and administrative staff.

Legal and Contractual Feasibility

The system complies with Ethiopian health data regulations and follows international best practices for medical data privacy and security. Patient data is encrypted both in transit and at rest, and access is restricted based on user roles.

Political Feasibility

The project aligns with national digital health initiatives and supports government efforts to expand healthcare access through technology-enabled solutions.

1.1.6 Significance of the Project

This project is significant in addressing critical gaps in dermatology service delivery and pharmaceutical access. By integrating consultation and medication procurement, the platform reduces patient travel, minimizes delays, and improves treatment continuity.

From an academic perspective, the project contributes to research on integrated digital healthcare systems. Practically, it provides a scalable model that can be adopted by clinics and healthcare institutions.

1.1.7 Beneficiaries of the Project

- **Patients:** Enhanced access to dermatology services, reduced costs, and improved convenience.
- **Dermatologists:** Efficient patient management and digital diagnostic tools.
- **Pharmacists:** Improved inventory visibility and prescription accuracy.
- **Administrators:** Better reporting, coordination, and operational control.

1.1.8 Scope of the Project

The project scope includes patient portals, appointment systems, payment processing, and pharmacy management. It focuses on digital service delivery while excluding physical logistics and insurance integration.

1.1.9 Methodology

The project adopts an Agile methodology supported by structured data gathering, iterative design, systematic implementation, and comprehensive testing to ensure reliability and usability.

1.1.10 Project Team and Communication

The multidisciplinary project team collaborates using modern communication tools to ensure transparency, accountability, and timely delivery of project milestones.

1.1.11 Development Tools and Technologies

A combination of industry-standard development, documentation, and collaboration tools is used to ensure code quality, maintainability, and efficient teamwork.

1.1.12 Testing and Deployment Environment

The system is tested in staging environments before deployment to production servers with redundancy, encryption, and backup mechanisms.

1.1.13 Constraints and Assumptions

The project assumes basic internet access and device availability while operating under strict timeline and bandwidth constraints.

1.1.14 Risks

Potential risks include data privacy breaches and internet instability, both of which are mitigated through encryption, access control, and system optimization.

1.1.15 Project Phases and Deliverables

The project follows a phased development approach, delivering incremental functionality and allowing for continuous evaluation and improvement.

1.1.16 Work Breakdown Structure

The WBS decomposes the project into manageable tasks covering setup, development, integration, testing, and deployment.

1.1.17 Project Schedule and Timeline

A detailed 12-week schedule ensures structured progress from design to deployment, with clearly defined milestones and deliverables.

Chapter 2

Business Area Analysis and Requirement Envisioning

2.1 Introduction

This chapter provides a thorough analysis of the business domain relevant to dermatology clinics and pharmacies, focusing on operational processes, stakeholder interactions, and existing limitations. The goal is to identify the gaps and inefficiencies in current systems and to envision a robust, web-based platform that integrates dermatology consultations with pharmacy services. This chapter will also define essential requirements for the system, both functional and non-functional, and present a conceptual domain model to guide system design.

2.1.1 Agile Analysis Approach

An Agile-based analysis methodology was adopted to ensure flexibility, rapid feedback incorporation, and stakeholder collaboration throughout the requirement gathering process. This approach allowed the project team to iteratively refine the system requirements while remaining adaptable to emerging insights. Key Agile analysis principles applied include:

- **Incremental Requirement Gathering:** Capturing high-priority features first (e.g., patient registration, appointment booking, prescription handling) and expanding based on iterative feedback.
- **Stakeholder Collaboration:** Continuous interaction with dermatologists, pharmacists, and patients to validate assumptions and refine workflows.
- **Adaptive Planning:** Adjusting requirements to accommodate legal, operational, and technical constraints discovered during early sprints.

- **Prioritization of Value:** Features were ranked based on impact on patient satisfaction, operational efficiency, and business growth.

2.1.2 Stakeholder Analysis

A detailed understanding of stakeholders is critical for designing a system that addresses their needs effectively. Table 2.1 summarizes the primary stakeholders and their roles:

Table 2.1: Stakeholder Roles and Responsibilities

Stakeholder	Responsibilities and Interests
Patient	Receives consultations, uploads medical images, books appointments, accesses prescriptions, pays fees, purchases medicines online.
Dermatologist	Reviews patient medical history, diagnoses skin conditions, issues prescriptions, monitors patient progress.
Pharmacist	Manages medicine inventory, verifies digital prescriptions, fulfills medicine orders, updates stock data.
Receptionist	Schedules appointments, manages patient registration, communicates reminders, maintains administrative records.
Administrator	Oversees system operations, monitors reports and analytics, manages user accounts, ensures compliance with legal requirements.
IT Support	Maintains the platform, ensures uptime, implements security patches, assists users with technical issues.

2.1.3 Players of the Existing System (Roles)

The existing healthcare system relies on several key actors:

- **Patient:** Visits the clinic physically, retains paper prescriptions, and coordinates separately with pharmacies.
- **Receptionist:** Manages manual registration, appointment scheduling, and paperwork.
- **Dermatologist:** Conducts consultations, records patient history manually, and issues handwritten prescriptions.
- **Pharmacist:** Dispenses medicines based on manual prescriptions and maintains stock records through notebooks or spreadsheets.

2.1.4 Current Activities and Workflow Analysis

Current workflows are largely manual and fragmented, leading to inefficiencies:

Patient Registration and Consultation

Patients fill out physical forms during clinic visits. The receptionist records details in ledger books, which may result in:

- Lost or incomplete patient data.
- Time-consuming retrieval of historical patient records.

Product Management and Stock Monitoring

Pharmacists manually record inventory and visually inspect stock levels. Limitations include:

- Difficulty in tracking expiration dates.
- No automated alerts for low stock levels.
- Increased risk of stockouts or selling expired medicine.

Purchase Orders and Replenishment

Purchase orders are communicated through phone calls or supplier visits. This leads to:

- Delays in procurement.
- Inefficient record-keeping of order history.
- Lack of accountability for suppliers.

Reporting and Communication

Reports are handwritten, and communication relies on phone calls or verbal instructions:

- High potential for errors in financial and inventory reporting.
- Missed appointments due to lack of automated reminders.
- Delayed communication with patients regarding prescriptions or test results.

2.1.5 Forms and Reports of the Existing System

Current documentation relies heavily on paper, including:

- Patient registration forms.
- Prescription sheets.
- Medicine stock cards and ledgers.
- Daily/monthly sales summaries.

2.1.6 SWOT Analysis of the Existing System

Table 2.2: SWOT Analysis of Current Dermatology and Pharmacy System

Aspect	Strengths	Weaknesses
Operational	Experienced staff and well-established clinic procedures	Manual workflows prone to errors, slow processing, lack of data integration
Financial	Low technological investment	High long-term operational costs due to inefficiency
Communication	Direct interaction with patients	No automated reminders, delayed communication, low patient satisfaction

Aspect	Opportunities	Threats
Technology	Adoption of integrated web-based solutions	Resistance to change, limited IT infrastructure in rural areas
Market	Expansion through online consultations and pharmacy services	Competition from emerging telemedicine platforms
Regulatory	Alignment with Ministry of Health digitalization initiatives	Compliance with data privacy and healthcare regulations

2.1.7 Problems of the Current System

Problems in Stock Entry and Inventory Management

High chance of human error in manual recording leads to:

- Duplicate or missing entries.
- Mismanagement of batch and expiration data.

Problems in Stock Monitoring

Manual inspections create challenges:

- Difficulty in maintaining accurate stock counts.
- Lack of automatic notifications for expiring or low-stock items.

Problems in Purchase Orders and Replenishment

Manual processes result in:

- Delayed fulfillment.
- Errors in order quantities.
- Inefficient coordination with suppliers.

Problems in Reporting

Handwritten reports are:

- Time-consuming.
- Error-prone.
- Lack real-time insight into patient or pharmacy operations.

Problems in Communication

- Missed appointment notifications.
- Delayed follow-ups on prescriptions or lab results.
- Low patient satisfaction and retention.

2.2 Requirements Envisioning

2.2.1 Non-Functional Requirements

- **Security:** Patient data encryption, secure login, multi-factor authentication.
- **Performance:** Low-latency system response even under high concurrent usage.
- **Scalability:** Ability to support additional clinics, users, and pharmacies.
- **Usability:** Accessible and intuitive interfaces for non-technical users.
- **Reliability:** High uptime with backup and recovery mechanisms.
- **Maintainability:** Modular codebase allowing future feature extensions.

2.2.2 Functional Requirements

- Patient profile management and registration.
- Appointment scheduling with dermatologists.
- Secure upload of medical images.
- Digital consultation and prescription issuance.
- Pharmacy inventory management and order fulfillment.
- Online payment processing.

- Automated notifications via SMS/email.
- Reporting dashboards for administrators.

2.2.3 System Requirements

Hardware: Modern computers, tablets, or smartphones for patients and staff; web servers for deployment.

Software: Modern browsers, database servers, backend servers, and secure APIs.

2.3 Essential Use Cases

2.3.1 Actor Specification

Patient: Registers, uploads images, books appointments, makes payments, and orders medicines.

Dermatologist: Reviews patient history, diagnoses, conducts consultations, issues prescriptions.

Pharmacist: Manages inventory, verifies prescriptions, processes orders.

Administrator: Manages users, monitors operations, generates reports.

2.3.2 Use Case Diagrams

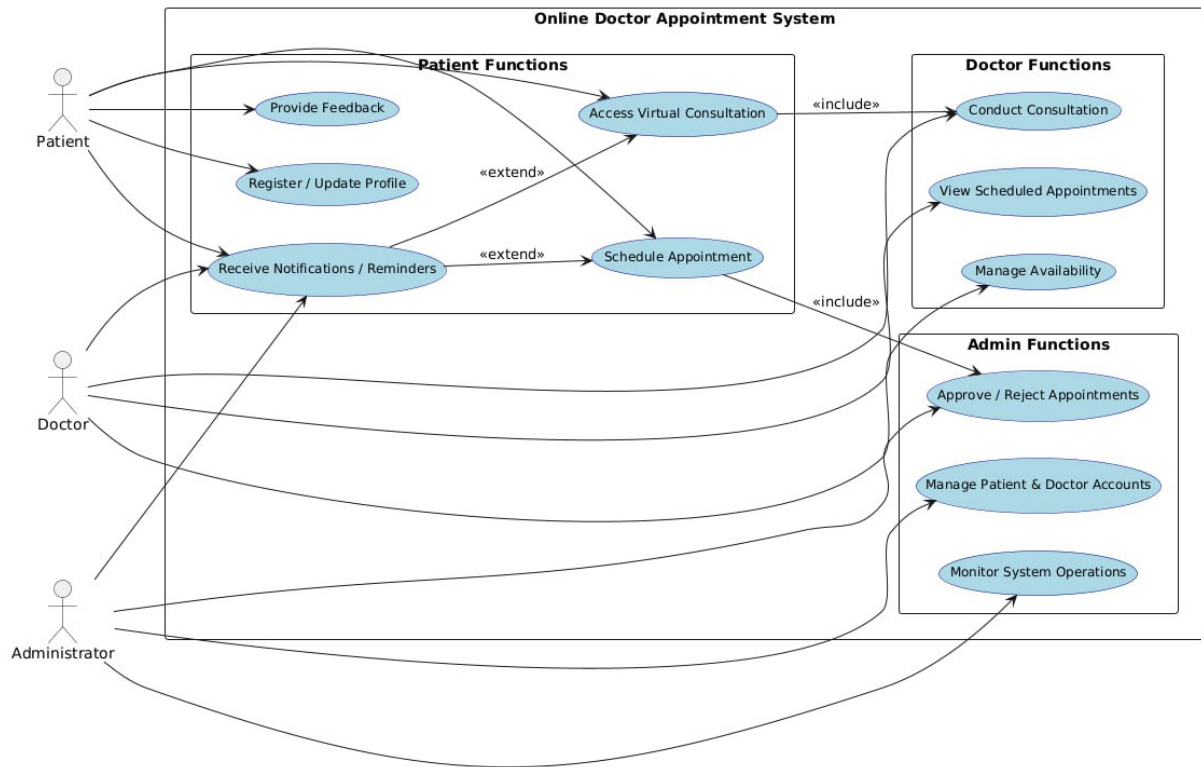


Figure 2.1: Overall Use-Case Diagram

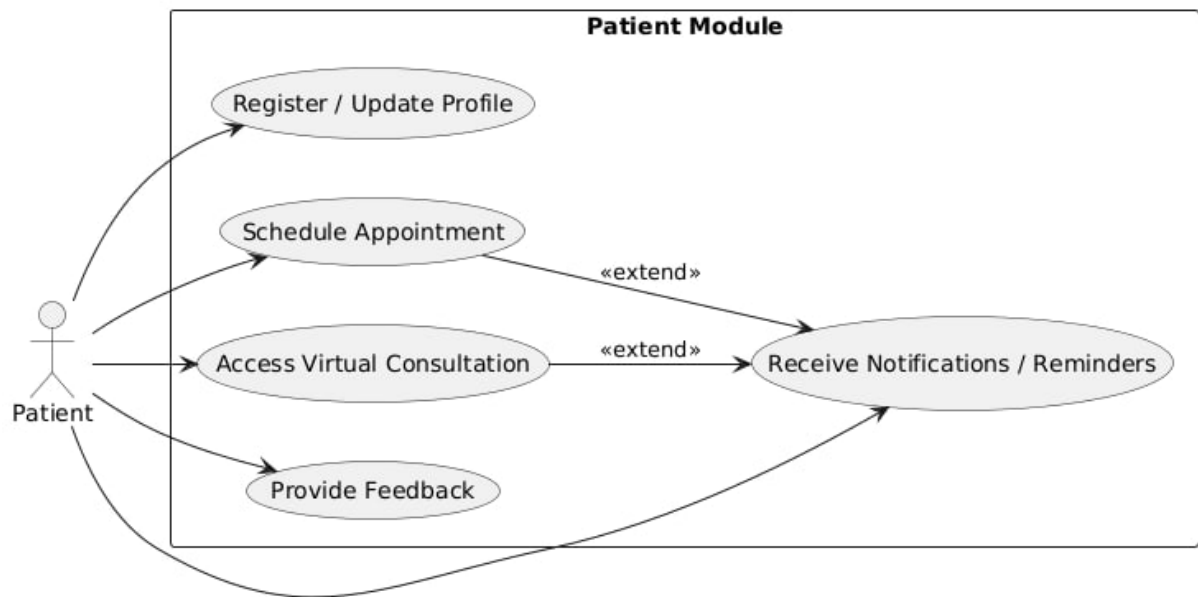


Figure 2.2: Patient Use Case Diagram

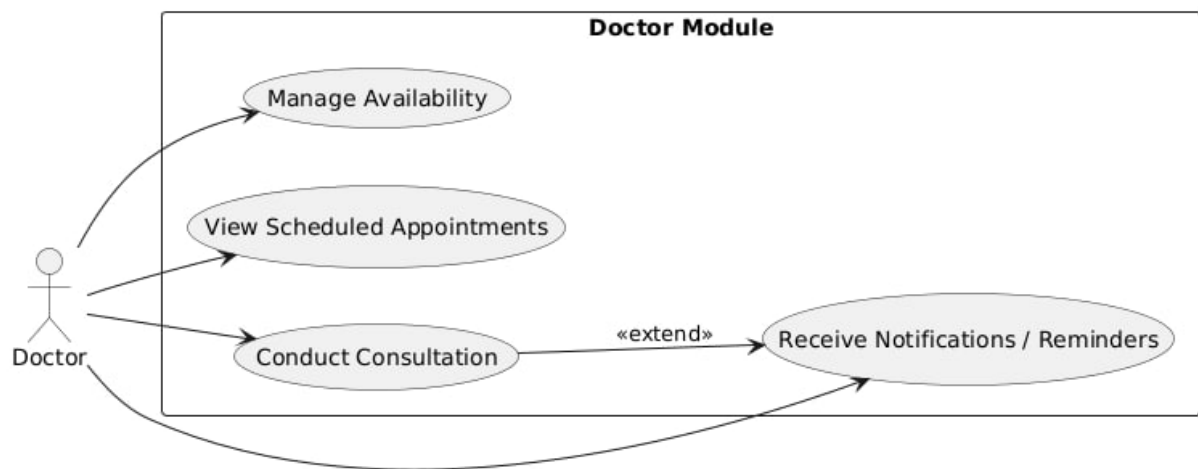


Figure 2.3: Doctor Use Case Diagram

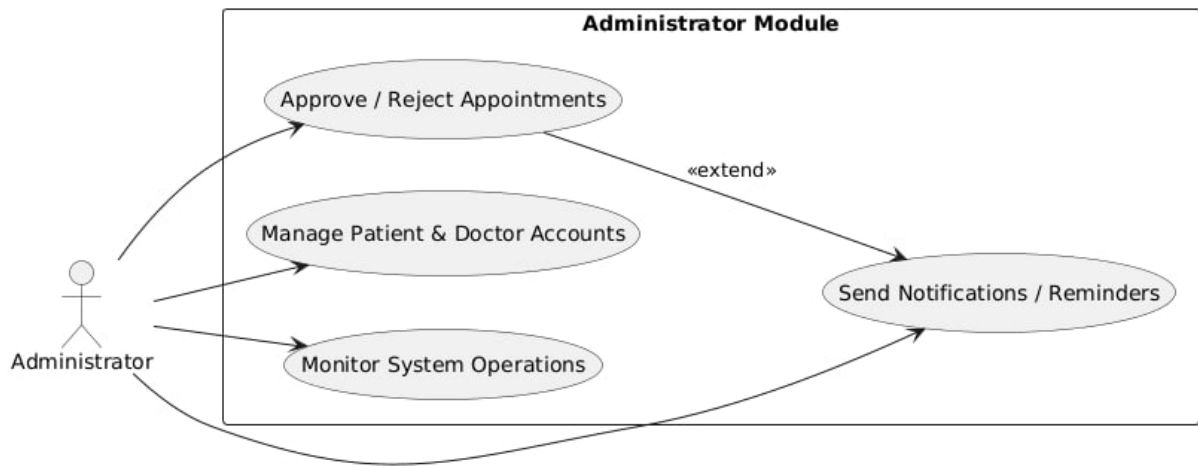


Figure 2.4: Administrator Use Case Diagram

2.4 Domain Modeling

Core entities include:

- **PatientProfile, Appointment, MedicalRecord, Prescription, Medicine, Payment.**

Relationships and attributes are defined to ensure relational integrity and effective workflow mapping.

2.5 Architectural Envisioning

2.5.1 System Architecture

A three-layer architecture is proposed:

- Presentation Layer (UI/UX)
- Application Layer (API/Business Logic)
- Data Layer (Database)

2.5.2 Deployment Diagram

- Web server delivers frontend assets.
- Application server handles API requests and business logic.
- Database server stores patient, prescription, and pharmacy data.

2.5.3 Persistence Data Management

PostgreSQL is employed to ensure strong consistency, transactional integrity, and reliable relational storage.

2.5.4 Access Control and Security

- JWT-based authentication.
- Role-based authorization.
- HTTPS for secure data transmission.
- Encrypted storage for sensitive patient records.

Chapter 3

Design and Construction

3.1 Introduction

This chapter focuses on the detailed design and construction of the proposed integrated dermatology and pharmacy web platform. The chapter covers actor identification, business rules, system decomposition, dynamic behavior via sequence diagrams, data modeling, architectural design, user interface design, and security considerations. The aim is to provide a comprehensive blueprint for implementation, ensuring that the system is modular, scalable, and maintainable.

The design approach follows standard software engineering principles, emphasizing ****modularity, separation of concerns, usability, security, and extensibility****. The chapter also integrates UML diagrams to provide a visual understanding of system components, interactions, and data flows.

3.2 Actor Identification

Actors are individuals or systems that interact with the platform. The correct identification of actors is crucial to defining system boundaries and responsibilities.

- **Patient:** End-user seeking dermatological consultation and medicine procurement. Responsibilities include registration, profile management, image upload, appointment booking, and online payment.
- **Dermatologist:** Provides diagnosis, manages patient medical records, issues prescriptions, and monitors treatment progress.
- **Pharmacist:** Manages inventory, verifies prescriptions, fulfills medicine orders, and updates stock levels.
- **System Administrator:** Oversees user management, monitors system performance, ensures compliance, and generates operational reports.

Additional system actors include:

- **Payment Gateway:** External system for processing card and mobile money payments.
- **Notification Service:** Handles SMS/email reminders for appointments and order updates.

3.3 Business Rules

Business rules define constraints and workflow policies for proper system operation. They ensure **consistency, security, and compliance**.

- Appointments must be paid for before confirmation.
- Medicine orders require a valid, authenticated prescription reference.
- Appointment reminders are automatically sent 24 hours prior via SMS/email.
- Stock levels must be updated in real time to prevent overbooking or over-selling of medicines.
- Each patient profile must have a unique identifier, and all transactions must be logged for traceability.
- Sensitive medical images must be encrypted at rest and in transit.

3.4 Sequence Diagrams

Sequence diagrams illustrate **dynamic interactions between actors and the system** over time. They help identify critical workflows and dependencies.

3.4.1 Appointment Booking Flow

1. Patient selects desired dermatology service and preferred time slot.
2. System checks dermatologists' availability in real-time.
3. Patient confirms the slot.
4. System redirects to the integrated payment gateway.
5. Payment is processed successfully.
6. System updates appointment status to "Confirmed" and triggers notification to both patient and dermatologist.

3.4.2 Pharmacy Order Flow

1. Patient uploads prescription or selects medicine from the digital prescription.
2. Pharmacist verifies prescription validity and stock availability.
3. Patient receives "Ready to Pay" alert.
4. Payment is processed through the gateway.
5. Order status is updated to "Processing," then "Ready for Pickup" or "In Transit."
6. Notification is sent to the patient.

3.5 System Decomposition

The platform is decomposed into **functional modules**, each responsible for a distinct subset of features. This modularization promotes **maintainability, scalability, and testability**.

3.5.1 Patient Module

Handles:

- User registration and authentication.
- Profile management and update of personal and medical data.
- Uploading images of skin conditions to cloud storage (AWS S3-compatible).
- Viewing consultation history and digital prescriptions.
- Tracking pharmacy orders.

3.5.2 Clinical Module

Handles:

- Virtual consultation interface using WebRTC/Google Meet APIs.
- Viewing patient history and uploaded images.
- Recording diagnoses and treatment notes in structured medical records.
- Generating digital prescriptions with automated links to the pharmacy module.

3.5.3 Pharmacy Inventory Module

Handles:

- Medicine catalog management with stock levels, batch numbers, and expiration dates.
- Barcode or QR-based inventory verification.
- Automated low-stock alerts.
- Linking prescriptions to medicine orders.

3.5.4 Payment and Billing Module

Handles:

- Integration with local mobile money providers and international card processors.
- Transaction status management.
- Generation of invoices and receipts.
- Secure storage of financial records for auditing.

3.6 Data Modeling

3.6.1 Entity-Relationship (ER) Diagram

Key entities and relationships:

- **PatientProfile** (1) — (N) **Appointment**
- **Appointment** (1) — (1) **Prescription**
- **Prescription** (1) — (N) **MedicineOrder**
- **Medicine** (1) — (N) **MedicineOrder**
- **Payment** (1) — (1) **Transaction**

3.6.2 Class Diagrams

Class diagrams define system objects, their attributes, and relationships:

- **PatientProfile:** patientID, name, email, phone, dateOfBirth, medicalHistory.
- **Appointment:** appointmentID, patientID, dermatologistID, date, time, status.
- **Prescription:** prescriptionID, appointmentID, medicineList, dosage, notes.
- **Medicine:** medicineID, name, batchNumber, quantity, expiryDate, price.
- **Payment:** paymentID, transactionID, amount, status, method.

3.6.3 Component Diagram

The component diagram shows ****high-level module interactions****:

- Patient Module interacts with Clinical Module and Pharmacy Module.
- Clinical Module interacts with Patient Module for data and with Pharmacy Module for prescriptions.
- Payment Module interfaces with external Payment Gateway.
- Notification Service interfaces with all modules for alerts.

3.7 User Interface Design

3.7.1 Design Principles

The platform adopts a ****mobile-first responsive design****, ensuring usability across smartphones, tablets, and desktops.

- **Accessibility:** Large interactive elements, screen reader compatibility, high-contrast color palettes.
- **Consistency:** Uniform layouts and typography across modules.
- **Feedback and Responsiveness:** Real-time alerts for booking confirmations, prescription availability, and payment status.
- **Security Manifest:** Masking sensitive inputs (passwords, payment details) and warning prompts for incorrect data entry.

3.7.2 Wireframe Overview

- **Patient Dashboard:** Upcoming appointments, recent prescriptions, order tracking.
- **Dermatologist Dashboard:** Patient queue, image submissions, consultation notes.
- **Pharmacy Dashboard:** Inventory alerts, prescription verification queue, order tracking.
- **Admin Dashboard:** User management, analytics, system logs.

3.8 Security Considerations

Security is central due to sensitive patient data and financial transactions:

- HTTPS for all web communications.
- JWT-based authentication for user sessions.
- Role-based access control to enforce least privilege.
- Encryption of all sensitive fields (medical history, prescriptions, payment data) in the database.
- Regular audit logs for compliance and traceability.

3.9 Design Rationale and Justification

The chosen design reflects:

- **Modularity:** Each module can be developed and tested independently.
- **Scalability:** Layered architecture and cloud storage allow future expansion.
- **Maintainability:** Standardized coding practices and ORM usage (Prisma) facilitate updates.
- **User-Centric Approach:** Dashboards and notifications designed for usability by medical and non-technical staff.
- **Integration Capability:** Designed for future integration with AI-assisted diagnosis or national health systems.

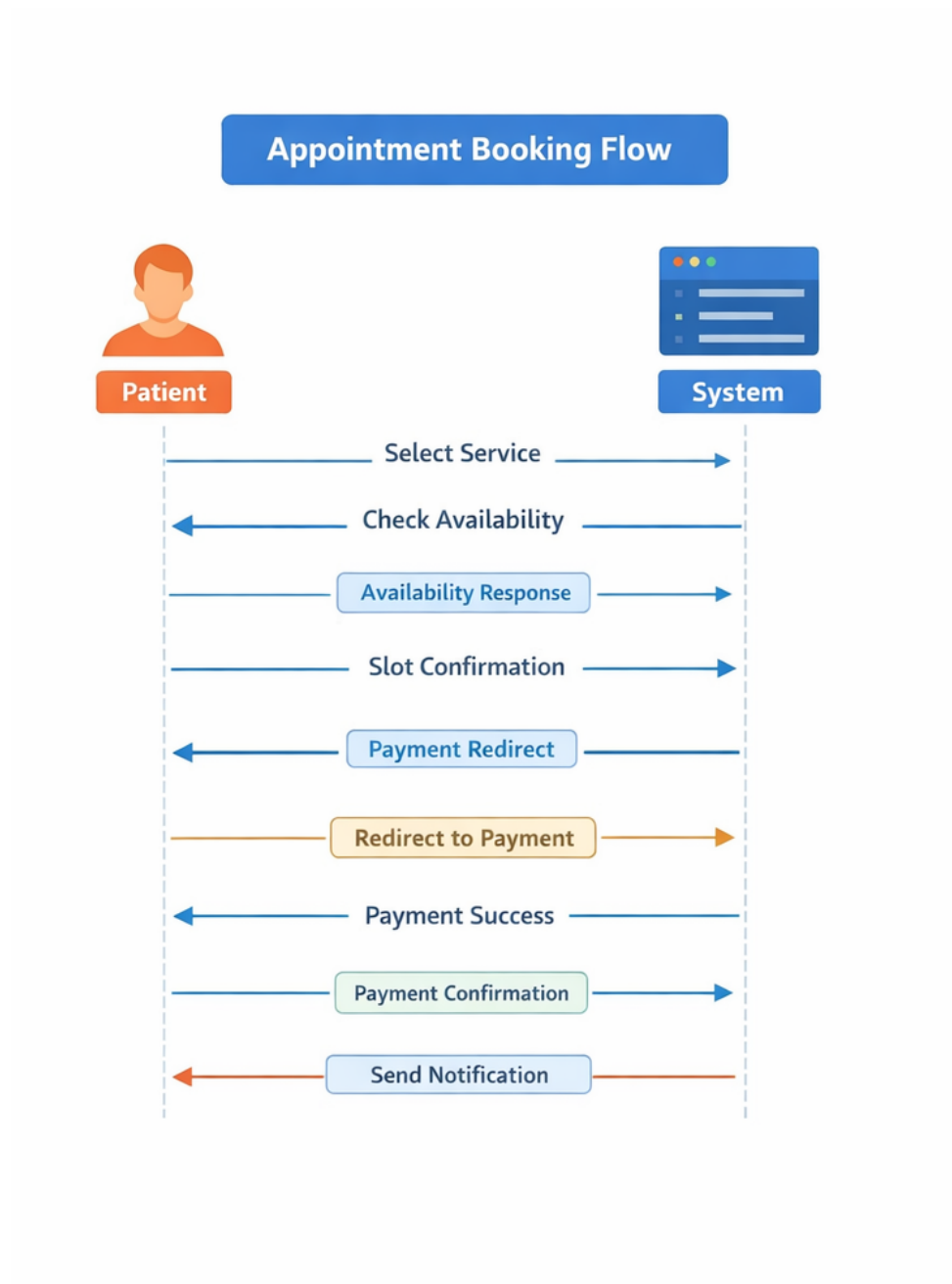


Figure 3.1: Sequence Diagram: Appointment Booking Flow

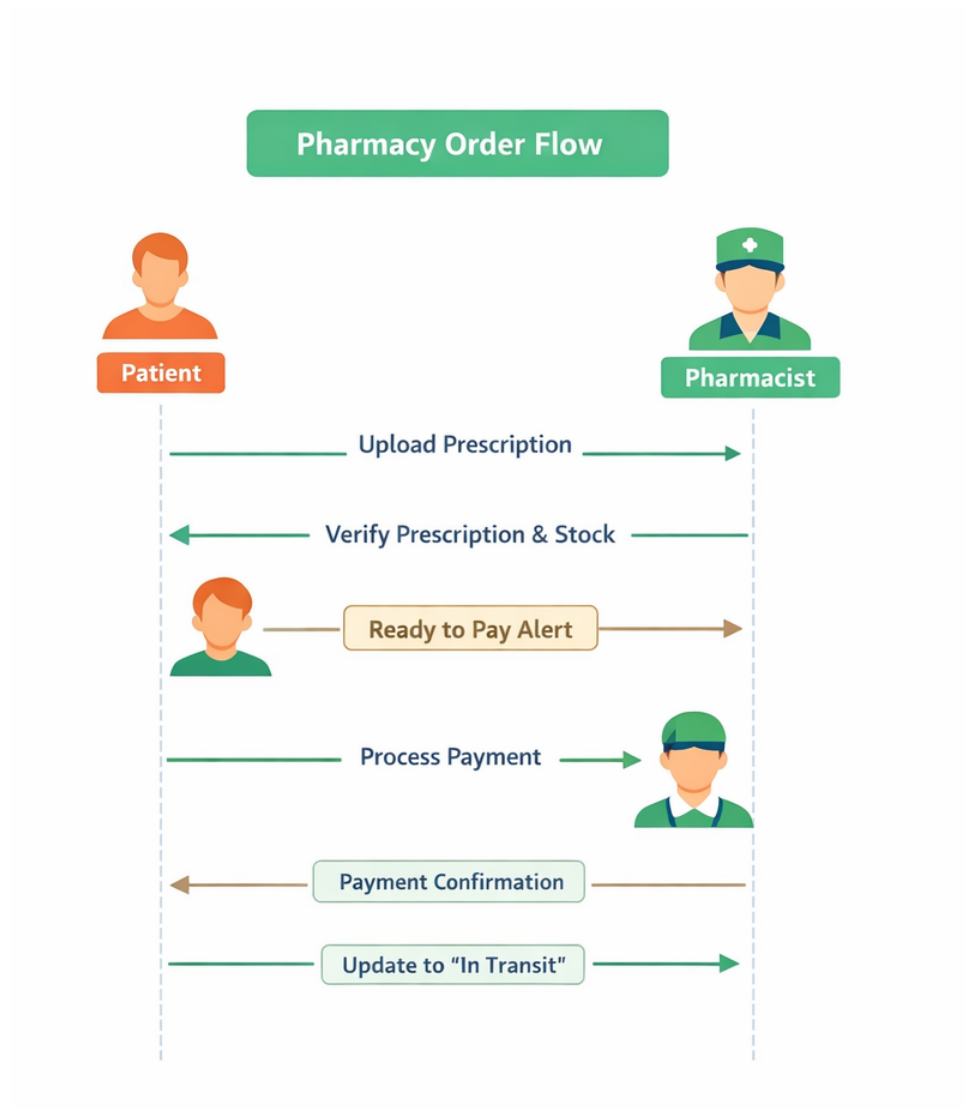


Figure 3.2: Sequence Diagram: Pharmacy Order Flow

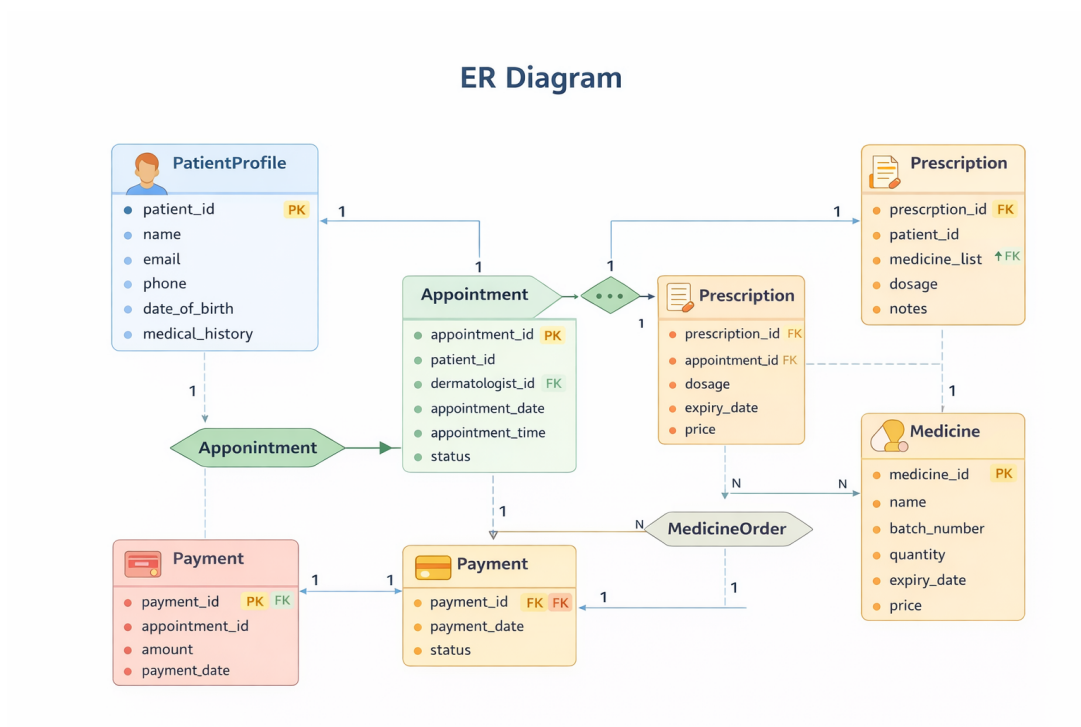


Figure 3.3: Entity-Relationship Diagram of the System

Chapter 4

Conclusion and Recommendations

4.1 Summary of Findings

The development of the "Online Dermatology Platform with Integrated Online Pharmacy" has demonstrated the feasibility and benefits of integrating specialized clinical services with pharmaceutical workflows. The platform effectively streamlines the patient journey from initial consultation to medication procurement, reducing the time and effort required for patients to access skincare services. By automating appointment scheduling, image management, and prescription handling, the system minimizes administrative errors and improves data accessibility for both patients and healthcare providers.

4.2 Achievement of Objectives

The project successfully met its predefined objectives:

- **Online Registration and Profile Management:** Patients can securely create and manage their digital health profiles.
- **Appointment Booking System:** A robust scheduling mechanism allows for selecting dermatology services and available slots.
- **Remote Consultation Facilitation:** Secure medical image uploads enable dermatologists to perform asynchronous diagnoses.
- **Secure Payment Gateway:** Integration with Chapa provides a reliable mechanism for consultation fees and medicine purchases.
- **Pharmacy Module:** A dedicated inventory management system allows pharmacists to track stock and fulfill orders efficiently.

4.3 Limitations

Despite the successful implementation, several limitations were identified:

- **Internet Connectivity:** The system's performance is highly dependent on stable internet access, which remains a challenge in certain regions of Ethiopia.
- **Physical Medicine Delivery:** While the platform handles orders and payments, the actual physical delivery of medication still requires a separate logistical infrastructure.
- **Integration with National HER:** The system currently operates as a standalone platform and is not yet integrated with national Electronic Health Record (EHR) systems.

4.4 Recommendations and Future Work

To further enhance the platform, the following recommendations are proposed:

- **AI-Assisted Diagnosis:** Incorporating machine learning models to assist dermatologists in identifying common skin conditions from uploaded images could improve diagnostic speed and accuracy.
- **Mobile Application:** Developing a native mobile application (Android/iOS) would provide a more seamless experience and enable push notifications for reminders.
- **Integration with Delivery Services:** Partnering with local delivery providers to automate the final leg of medicine procurement.
- **Blockchain for Prescriptions:** Implementing blockchain technology to ensure the immutability and security of digital prescriptions, preventing unauthorized modifications.

Chapter 5

References

1. Warshaw, E. M., et al. (2020). *Teledermatology: A Review and Update*. American Journal of Clinical Dermatology.
2. Ethiopia Ministry of Health. (2021). *Information Revolution: Digital Health Blueprint (2021-2030)*.
3. Whited, J. D. (2016). *Teledermatology: Care Models and Implementation*. Current Dermatology Reports.
4. Sagner, J. (2021). *The Growth of E-Pharmacy in Emerging Markets*. Journal of Pharmaceutical Policy and Practice.
5. Chapa. (2023). *Integration Guide for Ethiopian Fintech Solutions*.
6. American Academy of Dermatology (AAD). (2024). *Teledermatology Practice Standards and Guidelines*.
7. World Health Organization (WHO). (2023). *Digital Health Trends and Regulatory Frameworks for E-Pharmacy*.
8. Vercel. (2024). *Next.js Documentation: Server Actions and App Router Architecture*.
9. PostgreSQL Global Development Group. (2023). *PostgreSQL 16 Documentation: Relational Database Design and Optimization*.
10. Node.js Foundation. (2024). *Node.js v20 LTS Documentation: Scalable Backend Services with Express*.
11. Kanthraj, G. R. (2025). *Teledermatology: The New Era of Digitalization in Dermatology Care*. ResearchGate.

12. National Association of Boards of Pharmacy (NABP). (2024). *Internet Pharmacy Standards and Patient Safety Protocols*.
13. Prisma Team. (2024). *Prisma ORM documentation: Type-safe Database Access for TypeScript*.

Chapter 6

Appendix

6.1 Appendix A: User Interview Guide

The following questions were used during the requirement gathering phase with dermatologists and pharmacists:

1. How do you currently manage patient records and appointments?
2. What are the biggest challenges in the manual prescription process?
3. How frequently do you encounter issues with medicine stock availability tracking?
4. What features would be most beneficial for a remote consultation platform?
5. What are your primary concerns regarding data security and patient privacy?

6.2 Appendix B: Database Schema (Detailed)

Table: Users

- `id`: UUID (Primary Key)
- `email`: String (Unique)
- `password_hash`: String
- `role`: Enum (PATIENT, DOCTOR, PHARMACIST, ADMIN)
- `created_at`: Timestamp

Table: Appointments

- `id`: UUID (Primary Key)
- `patient_id`: UUID (FK to Users)

- **doctor_id:** UUID (FK to Users)
- **status:** Enum (PENDING, CONFIRMED, CANCELLED, COMPLETED)
- **scheduled_at:** Timestamp

Table: Medicines

- **id:** UUID (Primary Key)
- **name:** String
- **stock_quantity:** Integer
- **price:** Decimal
- **expiry_date:** Date

6.3 Appendix C: Project Timeline

The project was executed over a 12-week period following an Agile Scrum methodology:

Phase	Duration	Key Deliverables
Planning	Weeks 1-2	Project Charter, Feasibility Study, PRD
Analysis	Weeks 3-4	UML Diagrams, Wireframes, Database Schema
Development	Weeks 5-9	Frontend, Backend, API Integration, Payments
Testing	Weeks 10-11	Unit Testing, UAT, Security Audits
Deployment	Week 12	Production Launch, User Manual, Final Report

6.4 Appendix D: User Manual Fragments

How to Book an Appointment:

1. Login to the Patient Dashboard.
2. Click on "Book New Appointment".
3. Select a Dermatology Service and a preferred doctor.
4. Choose an available date and time slot.
5. Complete the registration payment via Chapa.

How to Fulfill a Prescription (Pharmacist):

1. Access the "Prescription Queue".
2. Select a pending prescription for verification.
3. Verify stock availability and batch numbers.
4. Mark the order as "Ready for Pickup" or "Out for Delivery".